MS 6015: Multivariate Statistical Methods for Business

Factor Analysis

Reference: Johnson and Wichern: Chapter 9

Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

Introduction

- Factor analysis is used to uncover the latent structure (dimensions) of a set of variables. It reduces attribute space from a larger number of variables to a smaller number of factors .
- The essential purpose of factor analysis is to describe, if possible, the covariance relationships among many variables in terms of a few underlying, but unobservable, random quantities called factors.
- Suppose variables can be grouped by their correlations. Variables in the same group are highly correlated while variables in different groups have relatively small correlations.

Introduction

- It is conceivable that each group of variables represents a single underlying construct, or factor, that is responsible for the observed correlations.
- For example, correlations from the group of test scores in classics, French, English, mathematics, and music collected by Spearman suggested an underlying "intelligence" factor.

Introduction

- Factor analysis can be considered an extension of principal component analysis. Both can be viewed as attempts to approximate the covariance matrix Σ. However, the approximation based on the factor analysis model is more elaborate.
- The primary question in factor analysis is whether the data are consistent with a prescribed structure.

Purpose and Estimation

- Purpose
  - Explore underlying structure - data driven.
  - Confirm underlying structure - theory driven.
- Estimation method (factor extraction)
  - Principal component method
  - Maximum Likelihood method

Exploratory factor analysis

- Exploratory factor analysis (EFA) seeks to uncover the underlying structure of a relatively large set of variables.
- The researcher's *àpriori* assumption is that any indicator may be associated with any factor.
- This is the most common form of factor analysis.
- There is no prior theory and one uses factor loadings to intuit the factor structure of the data.

Confirmatory factor analysis

- Confirmatory factor analysis (CFA) seeks to determine if the number of factors and the loadings of measured (indicator) variables on them conform to what is expected on the basis of pre-established theory.
- Indicator variables are selected on the basis of prior theory and factor analysis is used to see if they load as predicted on the expected number of factors.
- The researcher's *àpriori* assumption is that each factor (the number and labels of which may be specified *àpriori*) is associated with a specified subset of indicator variables.

Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

Spearman's Experiment

- The idea that general intelligence (g) exists comes from the work of Charles Spearman (1863-1945) who helped develop the factor analysis approach in statistics.
- Studied correlations between student test scores for various types and developed a simple model to account for the observed correlations.

Spearman's Experiment

	Cl	Fr	En	Ma	Di	Mu
Cl	**1.00**	.74	.69	.67	.63	.60
Fr	.74	**1.00**	.65	.62	.59	.56
En	.69	.65	**1.00**	.59	.55	.52
Ma	.67	.62	.59	**1.00**	.53	.50
Di	.63	.59	.55	.53	**1.00**	.48
Mu	.60	.56	.52	.50	.48	**1.00**

Can the correlations between test scores be explained by a single underlying factor?

Observations

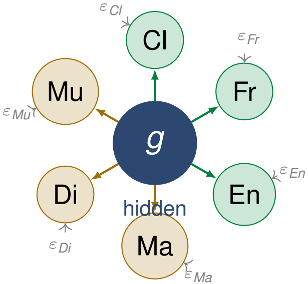
- Spearman developed a simple model to account for the observed correlations.
- He observed an interesting property of the matrix.

Observations

- Spearman developed a simple model to account for the observed correlations.
- He observed an interesting property of the matrix.
- Any two rows are almost proportional if the diagonals are ignored.

$$\frac{.83}{.67} \approx \frac{.70}{.64} \approx \frac{.66}{.54} \approx \frac{.63}{.51}$$

First impression Every pair of subjects is positively correlated. Even Classics and Music ($r = .63$). Something common must be driving all of them!



Key insight: g drives all 6 scores. Each score also has its own unique "noise" ε_i. The correlation between any two scores is entirely due to g.

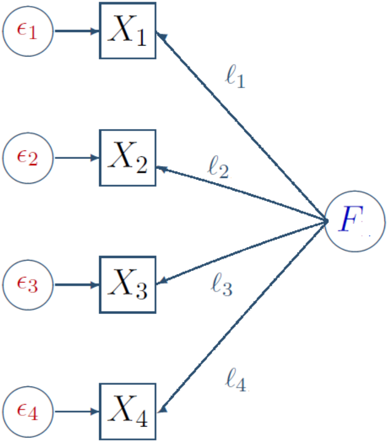
Observations

- Based on this observation, he suggested that the six test scores are described by a equation

$$X_i = l_i F + \epsilon_i$$

  where
  - X_i is the i^{th} score standardized to have mean 0 and standard deviation 1.
  - l_i is a constant
  - F is a factor which is a random variable with mean 0 and standard deviation 1.
  - ϵ_i : Part of X_i specific to the i^{th} test only.
  - F and ϵ_i are independent.
- He showed that the constant ratio between the rows of a correlation matrix follows this assumption.

One Common Factor Model



Why Does This Matter?

If **one hidden factor** g explains all correlations, then:

$$r_{ij} = l_i\, l_j \quad \Rightarrow \quad \frac{r_{ij}}{r_{kj}} = \frac{l_i\, \cancel{l_j}}{l_k\, \cancel{l_j}} = \frac{l_i}{l_k}$$

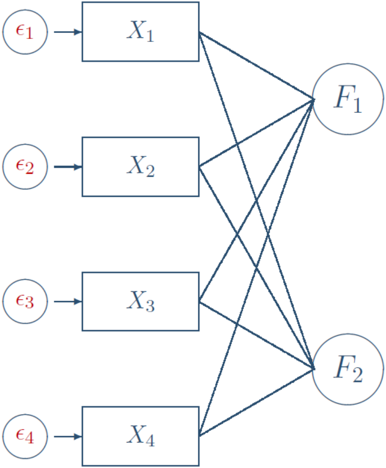
The ratio is **constant across** j ? exactly what we observe!

The One-Factor Model

$$X_i = l_i\, F + \varepsilon_i$$

- F: general intelligence g, with $\mathrm{Var}(F) = 1$
- l_i: loading of test i on g
- ε_i: unique (test-specific) part
- $F \perp \varepsilon_i; \quad \varepsilon_i \perp \varepsilon_j$

Two Factor Model



Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

PCA versus FA

- In PCA, the principal components are determined as linear combinations of original variables. In factor analysis, the original variables are expressed as linear combinations of factors.
- In PCA, we explain a large part of total variance. In factor analysis, we seek to account for covariances /correlations among variables

PCA versus FA

Principal Components	**Factor Analysis**
Descriptive Method for Data Reduction	Statistical Method which can be tested
Accounts for variance of Data	Accounts for pattern of correlations
Components are always uncorrelated	Factors may or may not be correlated
Components are linear combinations of observed variables	Factors are linear combinations of common parts of variables(unobservable)
Scores on components can be computed exactly	Scores on factors are estimated

The Orthogonal Factor Model

- The observable random vector X, with p components, has mean μ and covariance matrix Σ.
- The factor model postulates that X is linearly dependent upon a few unobservable random variables $F_1, F_2, \ldots, F_m$ called common factors, and p additional sources of variation $\epsilon_1, \epsilon_2, \ldots, \epsilon_p$ called errors, sometimes, specific factors.
- In other words, factor analysis expresses each variable as a linear combination of underlying common factors $F_1, \ldots, F_m$ with an error term to account for that part of the variable that is unique and not common to other variables.

Factor Analysis Model

$$\begin{array}{ccccccccccccc}
X_1 - \mu_1 & = & l_{11}F_1 & + & l_{12}F_2 & + & \ldots & + & l_{1m}F_m & + & \epsilon_1 \\
X_2 - \mu_2 & = & l_{21}F_1 & + & l_{22}F_2 & + & \ldots & + & l_{2m}F_m & + & \epsilon_2 \\
\vdots & & & & & & \vdots & & & & \\
X_p - \mu_p & = & l_{p1}F_1 & + & l_{p2}F_2 & + & \ldots & + & l_{pm}F_m & + & \epsilon_p
\end{array}$$

In matrix notation

$$\mathbf{X} - \mu = \mathbf{LF} + \varepsilon$$

- $F_1, \ldots, F_m$ are the common factors.
- l_{ij} is called the loading of the i^{th} variable on the j^{th} factor.
- $\mathbf{L}$ is the matrix of factor loadings
- ϵ_i are the specific factors.

Assumptions on F and ε

The unobservable random vectors **F** and ε satisfy the following conditions:

- **F** and ε are independent
- $E(\mathbf{F}) = 0, Cov(\mathbf{F}) = I$
- $E(\varepsilon) = 0, Cov(\varepsilon) = \Psi = \begin{bmatrix} \psi_1 & 0 & \dots & 0 \\ 0 & \psi_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \psi_p \end{bmatrix}$
- **F** and ε are independent, so $Cov(\mathbf{F}, \varepsilon) = E(\varepsilon \mathbf{F}^{'}) = 0$

Covariance structure for the Orthogonal Factor Model

The Orthogonal Factor Model implies a covariance structure for **X**.

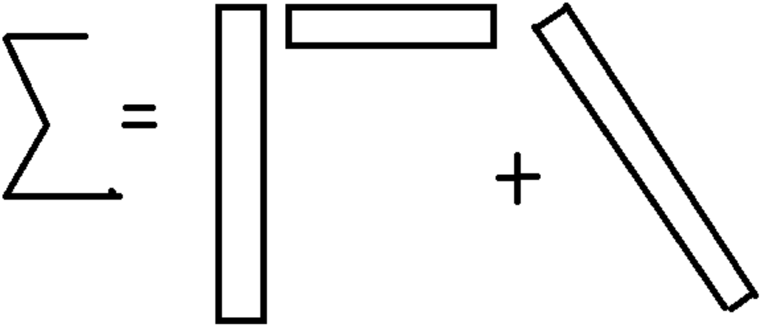
- $Cov(X) = LL' + \Psi$ or

$$Var(X_i) = l_{i1}^2 + l_{i2}^2 + \ldots + l_{im}^2 + \psi_i$$

$$Cov(X_i, X_k) = l_{i1} l_{k1} + \ldots l_{im} l_{km}$$

- Cov(X,F) = L or

$$Cov(X_i, F_j) = l_{ij}$$



Communality and Specific Variance

That portion of the variance of the i^{th} variable contributed by the m common factors is called the i^{th} communality. That portion of $Var(X_{ii}) = \sigma_{ii}$ due to the specific factor is often called the uniqueness, or specific variance. Denoting the i^{th} communality by h_i^2 , we have

$$\underbrace{\sigma_{ii}}_{Var(X_i)} = \underbrace{l_{i1}^2 + l_{i2}^2 + \ldots + l_{im}^2}_{Communality} + \underbrace{\psi_i}_{Specific Variance}$$

or $h_i^2 = l_{i1}^2 + l_{i2}^2 + \ldots + l_{im}^2$ and $\sigma_{ii} = h_i^2 + \psi_i; \quad i = 1, \ldots, p$
The i^{th} communality is the sum of squares of the loadings of the i^{th} variable on the m common factors.

Factor Model-Observations

- The diagonal elements of Σ can easily be modelled by adjusting Ψ so $LL^{'}$ is a simplified configuration of the off-diagonal elements.
- Hence, the critical aspect of the Factor Model involves covariances.
- If X has dimension p, then $\Sigma = LL^{'} + \Psi$ always holds for $m = p$. However, nothing has been gained in this case.
- The success of Factor Analysis depends on how well Σ can be expressed as $LL^{'} + \Psi$ with m small.

Example 1:Verifying the relation for two factors

Consider the covariance matrix

$$\Sigma = \begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix}$$

Example 1:Verifying the relation for two factors

By matrix algebra, we can verify the equality $\Sigma = \mathbf{L}\mathbf{L}^{'} + \Psi$

$$\begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 4 & 7 & -1 & 1 \\ 1 & 2 & 6 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Example 1:Verifying the relation for two factors

Therefore, Σ has the structure produced by an $m = 2$ orthogonal factor model.
Since the communality of X_i is $h_i^2 = l_{i1}^2 + l_{i1}^2 = 4^2 + 1^2 = 17$ the $Var(X_1)$ can be decomposed as

$$\begin{array}{ccccc}
\underbrace{\sigma_{11}} & = & \underbrace{l_{11}^2 + l_{12}^2} & + & \underbrace{\psi_1} \\
Var(X_1) & = & Communality & + & SpecificVariance \\
19 & = & 4^2 + 1^2 & + & 2
\end{array}$$

Example 2:

Unfortunately, for the factor analyst, most covariance matrices cannot be factored as $\Sigma = \mathbf{L}\mathbf{L}^{'} + \Psi$, where the number of factors m is much less than p.

Let $p = 3$ and $m = 1$ and suppose X_1, X_2 and X_3 have the covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.7 \\ 0.9 & 1 & 0.4 \\ 0.7 & 0.4 & 1 \end{bmatrix}$$

Show Σ can not be factored by a factor analysis model with $m = 1$.

Analysis of Factor Model

The analysis of the factor model proceeds by imposing conditions that allow one to uniquely estimate L and The loading matrix is then rotated (multiplied by an orthogonal matrix), where the rotation is determined by some “ease-ofinterpretation" criterion. Once the loadings and specific variances are obtained, factors are identified, and estimated values for the factors themselves (called factor scores) are frequently constructed.

Two important issues

- How to estimate the parameters in the model?
- How do we fix a rotation?

Methods of Estimation

- Factor analysis model based on a random sample $X_1, X_2, \ldots, X_n$ on p variables with sample covariance matrix S and unknown population covariance matrix Σ.
- To find an estimator $\hat{L}$ such that $S \approx \hat{L}\hat{L}' + \hat{\Psi}$
- S is estimator of unknown Σ.

Methods

- Principal Component Method
- Principal Factor Method
- Maximum Likelihood Method

Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

Principal Component Method

- The principal component factor analysis of the sample covariance matrix S is specified in terms of its eigenvalue-eigenvector pair $(\hat{\lambda}_1, \hat{e}_1), (\hat{\lambda}_2, \hat{e}_2), \ldots, (\hat{\lambda}_p, \hat{e}_p)$ where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \ldots, \hat{\lambda}_p$
- Let $m < p$ be the number of common factors. Then the matrix of estimated factor loadings $\tilde{l}_{ij}$ is given by

$$\tilde{L} = \left[\sqrt{\hat{\lambda}_1}\hat{e}_1 : \sqrt{\hat{\lambda}_2}\hat{e}_2 : \ldots : \sqrt{\hat{\lambda}_m}\hat{e}_m \right]$$

- The estimated specific variances are provided by the diagonal elements of the matrix

Principal Component Method

- The estimated specific variances are provided by the diagonal elements of the matrix $S - \tilde{L}\tilde{L}'$, so

$$\tilde{\psi} = \begin{pmatrix} \tilde{\psi}_1 & 0 & \dots & 0 \\ 0 & \tilde{\psi}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \tilde{\psi}_p \end{pmatrix}$$

- Communalities are estimated as $\tilde{h}_i^2 = \tilde{l}_{i1}^2 + \tilde{l}_{i2}^2 + \ldots + \tilde{l}_{im}^2$

Proportion of Total Sample Variance and Model Fit

- Proportion of the total sample variance due to the j^{th} factor is

$$\frac{\sum_{i=1}^{p} \tilde{l}_{ij}^2}{s_{11} + s_{22} + \ldots + s_{pp}} = \frac{\hat{\lambda}_j}{s_{11} + s_{22} + \ldots + s_{pp}}$$

- Assess how well the model fits by considering the error matrix

$$E = S - (\tilde{L}\tilde{L}' + \tilde{\Psi})$$

Example: Stock-Price Data

- The weekly rates of return for five stocks (JP Morgan, Citibank,Wells Fargo, Royal Dutch Shell, and ExxonMobil) listed on the New York Stock Exchange were determined for the period January 2004 through December 2005.
- The weekly rates of return are defined as (current week closing price-previous week closing price)/(previous week closing price), adjusted for stock splits and dividends.
- The observations in 103 successive weeks appear to be independently distributed, but the rates of return across stocks are correlated, because as one expects, stocks tend to move together in response to general economic conditions.

Example: Factor analysis of stock-price data

Consider example of stock price data of five firms collected for 103 days.

Variable	One-factor solution		Two-factor solution		
	Estimated factor loadings F_1	Specific variances $\tilde{\psi}_i = 1 - \tilde{h}_i^2$	Estimated factor loadings F_1	Estimated factor loadings F_2	Specific variances $\tilde{\psi}_i = 1 - \tilde{h}_i^2$
1. J P Morgan	.732	.46	.732	−.437	.27
2. Citibank	.831	.31	.831	−.280	.23
3. Wells Fargo	.726	.47	.726	−.374	.33
4. Royal Dutch Shell	.605	.63	.605	.694	.15
5. ExxonMobil	.563	.68	.563	.719	.17
Cumulative proportion of total (standardized) sample variance explained	.487		.487	.769	

Residual Matrix

$$\mathbf{R} - \tilde{\mathbf{L}}\tilde{\mathbf{L}}' - \tilde{\boldsymbol{\Psi}} = \begin{bmatrix} 0 & -.099 & -.185 & -.025 & .056 \\ -.099 & 0 & -.134 & .014 & -.054 \\ -.185 & -.134 & 0 & .003 & .006 \\ -.025 & .014 & .003 & 0 & -.156 \\ .056 & -.054 & .006 & -.156 & 0 \end{bmatrix}$$

The proportion of the total variance explained by the two-factor solution is appreciably larger than that for the one-factor solution. However, for $m = 2$, $\tilde{L}\tilde{L}'$ produces numbers that are, in general, larger than the sample correlations.

Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

Maximum Likelihood Method

- Maximum Likelihood Estimation requires that the data are sampled from a multivariate normal distribution.
  - This is a drawback of this method.
  - Data is often collected on a Likert scale, especially in the social sciences. Because a Likert scale is discrete and bounded, these data cannot be normally distributed.
- Using the Maximum Likelihood Estimation Method, we must assume that the data are independently sampled from a multivariate normal distribution with mean vector μ and variance-covariance matrix of the form: $\Sigma = LL^{'} + \Psi$

Maximum Likelihood Method

Assume $X_1, X_2, \ldots, X_n$ are iid $\sim N_p(\mu; \Sigma)$. Then, the likelihood is

$$L(\mu, \Sigma) = (2\pi)^{-np/2} |\Sigma|^{-n/2} exp\left\{ \frac{-1}{2} \sum_{i=1}^{n} (X_i - \mu)^{'} \Sigma^{-1} (X_i - \mu) \right\}$$

Under the orthogonal factor Model $\Sigma = LL^{'} + \Psi$. Then the likelihood becomes

$$L(\mu, L, \Psi) = (2\pi)^{-np/2} |LL^{'} + \Psi|^{-n/2} exp\left\{ \frac{1}{2} \sum_{i=1}^{n} (X_i - \mu)^{'} (LL^{'} + \Psi)^{-1} (X_i - \mu) \right\}$$

For unique factors, we can restrict $L^{'} \Psi^{-1} L = \Delta$ a diagonal matrix.

Maximum Likelihood Method

- With principal component solution, extraction of an additional factor does not effect the values already extracted; this is not the case with the MLE or the principal factor method.
- Both MLE and principal factor methods can run into problems (Heywood cases- when communality estimate is greater than 1 while using R, improper solutions) where the ψ want to be negative.
- You get different results if you use S or R (ie, not scale invariant); however, this is not the case for MLE:
- You can get a better fit via MLE.
- Statistical tests become possible.

Statistical Test of Number of common factors

The orthogonal factor model pre-specifies m, the number of the common factors. In practice, m is often unknown. Here we consider, for a given m, a statistical test to test whether such an m is appropriate. Presented in terms of statistical hypothesis as

$$\begin{array}{ll} H_0 : & \Sigma = LL^{'} + \Psi \\ H_1 : & \quad Otherwise \end{array}$$

Generalized Likelihood Test Statistic is

$$2Log\left(\frac{max\{L(\mu,\Sigma) : \mu,\Sigma\}}{max\{L(\mu,L,\Psi) : \mu,L,\Psi\}}\right)$$

With Appropriate refinement called Bartlett correction, the appropriate significance level α test is Reject H_0 when

$$[n-1-(2p+4m+5)/6]log\left(\frac{|\tilde{L}\tilde{L}^{'}+\tilde{\Psi}|}{|(n-1)/nS|}\right) > \chi^2_{((p-m)^2-p-m)/2}(\alpha)$$

Outline

Introduction

Development of Factor Analysis
- Spearman's Experiment
- PCA versus FA

The Orthogonal Factor Model

Methods of Estimation
- Principal Component Method
- Maximum Likelihood Method
- Examples: Stock Price Data

Factor Rotation

Factor Scores

Stock Price Data: MLE Solution

Variable	Maximum likelihood			Principal components		
	Estimated factor loadings		Specific variances	Estimated factor loadings		Specific variances
	F_1	F_2	$\hat{\psi}_i = 1 - \hat{h}_i^2$	F_1	F_2	$\tilde{\psi}_i = 1 - \tilde{h}_i^2$
1. J P Morgan	.115	.755	.42	.732	−.437	.27
2. Citibank	.322	.788	.27	.831	−.280	.23
3. Wells Fargo	.182	.652	.54	.726	−.374	.33
4. Royal Dutch Shell	1.000	−.000	.00	.605	.694	.15
5. Texaco	.683	−.032	.53	.563	.719	.17
Cumulative proportion of total (standardized) sample variance explained	.323	.647		.487	.769	

Factor loadings **L** are determined only up to an orthogonal matrix **T**

Factor loadings **L** are determined only up to an orthogonal matrix **T**

- When $m > 1$, there is always some inherent ambiguity associated with the factor model.

Factor loadings $\mathbf{L}$ are determined only up to an orthogonal matrix $\mathbf{T}$

- When $m > 1$, there is always some inherent ambiguity associated with the factor model.
- Too see this, let $\mathbf{T}$ be any $m \times m$ orthogonal matrix, so that $\mathbf{T}^{'}\mathbf{T} = \mathbf{T}\mathbf{T}^{'} = I$

Factor loadings **L** are determined only up to an orthogonal matrix **T**

- When $m > 1$, there is always some inherent ambiguity associated with the factor model.
- Too see this, let **T** be any $m \times m$ orthogonal matrix, so that $\mathbf{T}'\mathbf{T} = \mathbf{T}\mathbf{T}' = I$
- $\mathbf{X} - \mu = \mathbf{L}\mathbf{F} + \varepsilon = \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{F} + \varepsilon = \mathbf{L}^*\mathbf{F}^* + \varepsilon$

Factor loadings $\mathbf{L}$ are determined only up to an orthogonal matrix $\mathbf{T}$

- When $m > 1$, there is always some inherent ambiguity associated with the factor model.
- Too see this, let $\mathbf{T}$ be any $m \times m$ orthogonal matrix, so that $\mathbf{T}'\mathbf{T} = \mathbf{T}\mathbf{T}' = I$
- $\mathbf{X} - \mu = \mathbf{LF} + \varepsilon = \mathbf{LTT}'\mathbf{F} + \varepsilon = \mathbf{L}^*\mathbf{F}^* + \varepsilon$
- The loadings $\mathbf{L}^* = \mathbf{LT}$ and $\mathbf{L}$ both give the same representation- We cannot distinguish loadings F^* from F based on observations of X; they both generate the same covariance matrix.

Factor loadings $\mathbf{L}$ are determined only up to an orthogonal matrix $\mathbf{T}$

- When $m > 1$, there is always some inherent ambiguity associated with the factor model.
- Too see this, let $\mathbf{T}$ be any $m \times m$ orthogonal matrix, so that $\mathbf{T}'\mathbf{T} = \mathbf{T}\mathbf{T}' = I$
- $\mathbf{X} - \mu = \mathbf{L}\mathbf{F} + \varepsilon = \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{F} + \varepsilon = \mathbf{L}^*\mathbf{F}^* + \varepsilon$
- The loadings $\mathbf{L}^* = \mathbf{L}\mathbf{T}$ and $\mathbf{L}$ both give the same representation- We cannot distinguish loadings F^* from F based on observations of X; they both generate the same covariance matrix.
- The communalities, given by the diagonal elements of $\mathbf{L}'\mathbf{L} = (\mathbf{L}^*)'(\mathbf{L}^*)$ are also unaffected by the choice of $\mathbf{T}$.

- Recall that multiplying by an orthogonal matrix rotates variables.
- This indifference to rotation of factors is used to improve interpretation of factors.

Graphical Approach

- If there are only two factors $m = 2$, we can use plots to determine a reasonable rotation.
- In this case, the rows of $\hat{\Lambda}$ are pairs $(\hat{\lambda}_{i1}.\hat{\lambda}_{i2})$ corresponding to $X_i, i = 1, 2, \ldots, p$.
- We choose an angle of rotation ϕ for the axes to move them closer to the groupings of points.
- We set

$$T = \begin{bmatrix} cos\phi & -sin\phi \\ sin\phi & cos\phi \end{bmatrix}$$

for the corresponding rotation matrix.

Factor Rotation

- Factor loadings obtained from the initial loadings by an orthogonal transformation have the same ability to reproduce the covariance (or correlation) matrix.
- An orthogonal transformation of the factor loadings, as well as the implied orthogonal transformation of the factors, is called factor rotation.
- If $\hat{L}$ is $p \times m$ matrix of estimated factor loadings the

$$L^* = \hat{L}T; where \;\; TT^{'} = T^{'}T = I$$

is a $p \times m$ matrix of “rotated" loadings.

Example

Data from a survey of a 12 year old girl on her acquaintances

Person	Kind	Intelligent	Happy	Likeable	Just
Friend 1	1	5	5	1	1
Sister	8	9	7	9	8
Friend 2	9	8	9	9	8
Father	9	9	9	9	9
Teacher	1	9	1	1	9
Male Friend	9	7	7	7	9
Friend 3	9	7	9	9	7

Example: Correlation Matrix

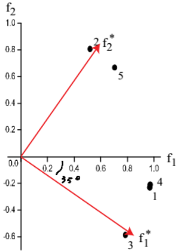
$$\mathbf{R} = \begin{pmatrix} 1 & .296 & \mathbf{.881} & \mathbf{.995} & .545 \\ .296 & 1 & -.022 & .326 & \mathbf{.837} \\ \mathbf{.881} & -.022 & 1 & \mathbf{.867} & .130 \\ \mathbf{.995} & .326 & \mathbf{.867} & 1 & .544 \\ .545 & \mathbf{.837} & .130 & .544 & 1 \end{pmatrix}$$

Example: Factor Loadings

Variables	Loadings		Communalities $\hat{h}_i^2$	Specific Variances $\hat{\psi}_i$
	$\hat{\lambda}_{1j}$	$\hat{\lambda}_{2j}$		
Kind	.969	-.231	.993	.007
Intelligent	.519	.807	.921	.079
Happy	.785	-.587	.960	.040
Likeable	.971	-.210	.987	.013
Just	.704	.667	.940	.060
Variance Accounted For	3.263	1.538	4.802	
Proportion of Total Variance	.653	.308	.960	
Cumulative Proportion	.653	.960	.960	

Observations

The factor loadings do not indicate an interpretation consistent with two groupings of variables, as the pattern of correlations in R indicated. An orthogonal rotation of -35^0 brings the axes closer to two clusters of points.



Example: Rotated Factor Loadings

Variables	Principal Component Loadings $\hat{\lambda}_{1j}$	Principal Component Loadings $\hat{\lambda}_{2j}$	Graphically Rotated Loadings $\hat{\lambda}^*_{1j}$	Graphically Rotated Loadings $\hat{\lambda}^*_{2j}$	Communalities $\hat{h}_i^2$
Kind	.969	-.231	**.927**	.367	.993
Intelligent	.519	.807	-.037	**.959**	.921
Happy	.785	-.587	**.980**	-.031	.960
Likeable	.971	-.210	**.916**	.385	.987
Just	.704	.667	.194	**.950**	.940
Variance Accounted For	3.263	1.538	2.696	2.106	4.802
Proportion of Total Variance	.653	.308	.539	.421	.960
Cumulative Proportion	.653	.960	.539	.960	.960

Factor Rotation

- The orthogonal factor model is not identifiable up to a rotation of the common factors or factor loading matrix. In other words,

$$X - \mu = LF + \epsilon = L^* F^* + \epsilon$$

  Therefore, up to a rotation, it is legitimate and often desirable to choose a pair $(L; F)$ so that it may achieve better interpretability.
- A pattern of loadings such that each variable loads highly on a single factor and has small to moderate loadings on the remaining factors.

Example

Lawley and Maxwell present the sample correlation matrix of examination scores in $p = 6$ subject areas for $n = 220$ male students. The correlation matrix is

$$
\begin{array}{cc}
 & \begin{array}{cccccc} \text{Gaelic} & \text{English} & \text{History} & \text{Arithmetic} & \text{Algebra} & \text{Geometry} \end{array} \\
\mathbf{R} = & \begin{bmatrix}
1.0 & .439 & .410 & .288 & .329 & .248 \\
 & 1.0 & .351 & .354 & .320 & .329 \\
 & & 1.0 & .164 & .190 & .181 \\
 & & & 1.0 & .595 & .470 \\
 & & & & 1.0 & .464 \\
 & & & & & 1.0
\end{bmatrix}
\end{array}
$$

Example: MLE Solution

A maximum likelihood solution for common factors yields the estimates

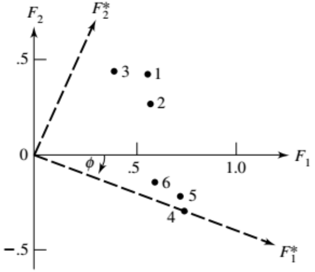
Variable	Estimated factor loadings F_1	F_2	Communalities $\hat{h}_i^2$
1. Gaelic	.553	.429	.490
2. English	.568	.288	.406
3. History	.392	.450	.356
4. Arithmetic	.740	−.273	.623
5. Algebra	.724	−.211	.569
6. Geometry	.595	−.132	.372

Example: Observations

- All the variables have positive loadings on the first factor.
  - Perhaps this factor reflects the overall response of the students to instruction and might be labeled a general intelligence factor.
- Half the loadings are positive and half are negative on the second factor.
  - A factor with this pattern of loadings is called a bipolar(“math-nonmath")factor.

Example: Factor Rotation for Test Scores

One new axis would pass through the cluster and the other through the group. Oblique rotations are so named because they correspond to a nonrigid rotation of coordinate axes leading to new axes that are not perpendicular.



Example: Estimated Rotated Factor Loadings

Variable	Estimated rotated factor loadings F_1^*	F_2^*	Communalities $\hat{h}_i^{*2} = \hat{h}_i^2$
1. Gaelic	.369	.594	.490
2. English	.433	.467	.406
3. History	.211	.558	.356
4. Arithmetic	.789	.001	.623
5. Algebra	.752	.054	.568
6. Geometry	.604	.083	.372

Varimax

- Kaiser has suggested an analytical measure of simple structure known as the varimax (or normal varimax) criterion.
- Let $\hat{L}^*$ be the $p \times m$ rotated factor loading matrix with elements $\hat{l}_{ij}^*$. Define $\tilde{l}_{ij}^* = \hat{l}_{ij}^* / \hat{h}_i$ and

$$V = \frac{1}{p} \sum_{j=1}^{m} \left[\sum_{i=1}^{p} (\tilde{l}_{ij}^*)^4 - \frac{1}{p} \left(\sum_{i=1}^{p} (\tilde{l}_{ij}^*)^2 \right)^2 \right]$$

  which is the sum of the column-wise variance of the squares of scaled factor loadings.
- Find optimal l_{ij}^* such the V achieves maximum.

Person Related data

Variables	Principal Component Loadings $\hat{\lambda}_{1j}$	Principal Component Loadings $\hat{\lambda}_{2j}$	Graphically Rotated Loadings $\hat{\lambda}^*_{1j}$	Graphically Rotated Loadings $\hat{\lambda}^*_{2j}$	Varimax Rotate Loadings $\hat{\lambda}^*_{1j}$	Varimax Rotate Loadings $\hat{\lambda}^*_{2j}$	Communalities $\hat{h}_i^2$
Kind	.969	-.231	**.927**	.367	**.951**	.298	.993
Intelligent	.519	.807	-.037	**.959**	.033	**.959**	.921
Happy	.785	-.587	**.980**	-.031	**.975**	-.103	.960
Likeable	.971	-.210	**.916**	.385	**.941**	.317	.987
Just	.704	.667	.194	**.950**	.263	**.933**	.940
Variance Accounted For	3.263	1.538	2.696	2.106	2.811	1.99 1	4.802
Proportion of Total Variance	.653	.308	.539	.421	.562	.398	.960
Cumulative Proportion	.653	.960	.539	.960	.562	.960	.960

Example: Consumer Preference

In a consumer-preference study, a random sample of customers were asked to rate several attributes of a new product. The response, on a 7-point semantic differential scale, were tabulated and the attribute correlation matrix constructed.

Data Set

Taste	Good buy for money	Flavor	Suitable for snack	Provides lots of energy
1.00	0.02	0.96	0.42	0.01
0.02	1.00	0.13	0.71	0.85
0.96	0.13	1.00	0.50	0.11
0.42	0.71	0.50	1.00	0.79
0.01	0.85	0.11	0.79	1.00

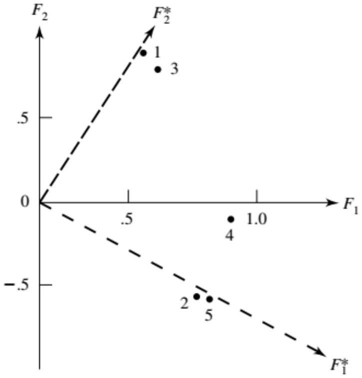
PCA Solution

Variable	Estimated factor loadings		Communalities	Specific variances
1. Taste	0.56	0.82	0.98	0.02
2. Good buy for money	0.78	-0.53	0.88	0.12
3. Flavor	0.65	0.75	0.98	0.02
4. Suitable for snack	0.94	-0.11	0.89	0.11
5. Provides lots of energy	0.80	-0.54	0.93	0.07
Eigenvalues	2.85	1.81		
Cumulative proportion of total (standardized) sample variance	0.571			

Observations

- It is clear that variables 2, 4, and 5 define factor 1 (high loadings on factor 1, small or negligible loadings on factor 2), while variables 1 and 3 define factor 2 (high loadings on factor 2, small or negligible loadings on factor 1).
- Variable 4 is most closely aligned with factor 1, although it has aspects of the trait represented by factor 2.
- We might call factor 1 a nutritional factor and factor 2 a taste factor.

Factor Loadings



Rotated Solution

Variable	Estimated factor loadings		Rotated estimated factor loadings		Cumumunalities
1. Taste	0.56	0.82	0.02	0.99	0.98
2. Good buy for money	0.78	-0.52	0.94	-0.01	0.88
3. Flavor	0.65	0.75	0.13	0.98	0.98
4. Suitable for snack	0.94	-0.1	0.84	0.43	0.89
5. Provides lots of energy	0.80	-0.54	0.97	-0.02	0.93
Cumulative proportion of total (standardized) sample variance	0.571	0.932	0.507	0.932	

Factor Scores

- The estimate values of the common factors, called factor scores may also required. These quantities are often used for diagnostic purposes, as well as inputs to a subsequent analysis.
- Factor scores are not estimates of unknown parameters in the usual sense. Rather, they are estimates of values for the unobserved random factor vectors $F_j, j = 1, \ldots, n$.
- That is, factor scores

$$\hat{f}_j = \text{estimate of the values } f_j \text{ attained by } F_j$$

Factor Score

- The estimation situation is complicated by the fact that the unobserved quantities f_j and ϵ_j outnumber the observed x_j.
- Two approaches to the problem of estimating factor values have been advanced. They are
  1. The Weighted Least Squares Method
  2. The Regression Method
- Both of the factor score approaches have two elements in common:
  1. They treat the estimate factor loadings $\hat{l}_{ij}$ and specific variance $\hat{\psi}_i$ as if they were true values.
  2. They involve linear transformations of the original data, perhaps centered or standardized. Typically, the estimated rotated loadings, rather than the original estimated loadings, are used to compute factor scores.

General Guidelines

To perform a complete factor analysis, some guidelines are useful. The following steps are recommended.

1. Perform principal component factor analysis, with care of the issue of standardization.
2. Perform a maximum likelihood factor analysis.
3. Compare the results of step 1 and step 2.
4. Change number of common factors m and repeat steps 1-3.
5. For large set of data, split them in half, perform the above analysis on each half and compare the results.